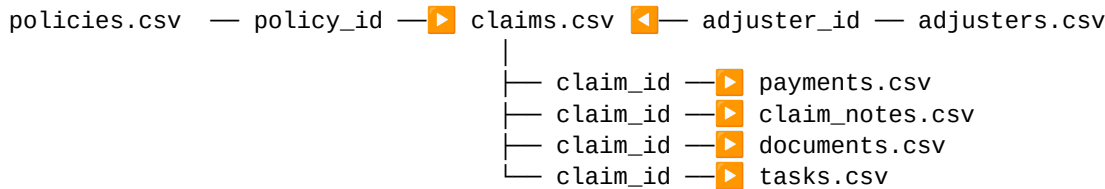


Synthetic Claims System CSV Data Dictionary

Multinational auto insurance prototype: USA, Australia, and UK

1. How the CSV Files Work Together



Step	CSV involved	What happens
1. Policy base	policies.csv	Represents customers, vehicles, coverage type, limits, deductibles, policy dates, and active/expired/cancelled status.
2. Claim event	claims.csv	Creates a claim linked to a policy and customer. Stores loss cause, status, geography, severity, fraud risk, and coverage validity.
3. Assignment	adjusters.csv + claims.csv	Claims are assigned to an adjuster, preferably in the same country/state or same country.
4. Review work	claim_notes.csv, documents.csv, tasks.csv	Stores adjuster notes, uploaded document metadata, and operational tasks needed to move a claim forward.
5. Settlement	payments.csv	Approved/paid/closed covered claims get payment records. Denied or invalid-coverage claims typically have no payment.
6. Reporting	All CSVs	Dashboards can aggregate claims by country, status, adjuster workload, fraud risk, severity, paid amount, and open tasks.

2. CSV File Summary

CSV file	Primary key	Connects by	Purpose
policies.csv	policy_id	policy_id -> claims.csv	Master policy and customer coverage data.
claims.csv	claim_id	policy_id, adjuster_id, claim_id	Central claim transaction file.
adjusters.csv	adjuster_id	adjuster_id -> claims/tasks/notes	Claim handler roster and territory/specialty details.
payments.csv	payment_id	claim_id -> claims.csv	Claim payment and settlement details.
claim_notes.csv	note_id	claim_id, adjuster_id	Adjuster activity log and claim history.
documents.csv	document_id	claim_id	Metadata for claim documents and evidence.
tasks.csv	task_id	claim_id, assigned_to	Operational checklist and work queue.

policies.csv

Purpose: The source of policy, customer, coverage, geography, and vehicle context.

Column	Meaning	Used for
policy_id	Unique identifier for each insurance policy.	Join to claims.csv to know which policy a claim belongs to.
customer_id	Synthetic customer identifier.	Group claims by customer or find repeat claimants.
customer_name	Synthetic customer name.	Display customer context in claim detail screens.
country	Country where the policy is issued: USA, Australia, or UK.	Country-level dashboards and filters.
state	State/region under the country, such as TX, NSW, England.	Regional reporting and adjuster matching.
policy_status	Active, Expired, or Cancelled.	Coverage validation and denial candidates.
policy_start_date	Policy effective start date.	Checks whether claim date falls inside coverage period.
policy_end_date	Policy expiration date.	Checks whether claim date falls inside coverage period.
coverage_type	Liability Only, Standard, or Full Coverage.	Segment claims by product/coverage level.
deductible	Amount customer pays before insurer payment.	Payment calculation.
coverage_limit	Maximum amount insurer may pay.	Exposure, reserve, and settlement calculations.
vehicle_year	Synthetic vehicle year.	Vehicle context and analytics.
vehicle_make	Synthetic vehicle make.	Vehicle context and analytics.
vehicle_model	Synthetic vehicle category/model.	Vehicle context and analytics.

claims.csv

Purpose: The core table of the prototype. Most dashboards start here.

Column	Meaning	Used for
claim_id	Unique identifier for each claim.	Primary record for claim screens and joins.
policy_id	Policy linked to the claim.	Join back to policy/customer/coverage details.
customer_id	Customer linked to the claim.	Customer-level reporting or repeat-claim analysis.
country	Country copied from the policy.	Country-level claim volume, cost, and risk reports.
state	State/region copied from the policy.	Regional claim trend reporting.
claim_date	Date of loss/report in the synthetic dataset.	Aging, trend charts, and timeline filters.
claim_type	Collision, Comprehensive, Bodily Injury, or Property Damage.	Operational segmentation and adjuster specialization.
loss_cause	Synthetic description of what happened.	Claim detail page and analysis by cause.
claim_status	New, Under Review, Approved, Denied, Paid, or Closed.	Workflow funnel and operational dashboards.
estimated_loss	Estimated claim cost before deductible and limit rules.	Financial exposure and severity reporting.
deductible	Deductible copied from policy.	Payment calculation and financial reporting.
coverage_limit	Limit copied from policy.	Caps maximum potential payment.
coverage_valid	True/False based on active policy and date window.	Coverage checks and denial candidates.
severity_score	0-100 synthetic score based mostly on estimated loss.	Triage and high-severity queues.
fraud_risk_score	0-100 synthetic score with higher values indicating more risk.	Fraud queue and executive risk reporting.
adjuster_id	Assigned claim handler.	Join to adjusters.csv and workload

		reporting.
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adjusters.csv

Purpose: The workforce table used for assignment and capacity reporting.

Column	Meaning	Used for
adjuster_id	Unique identifier for an adjuster.	Join to claims, notes, and tasks.
adjuster_name	Synthetic adjuster name.	Display assigned owner in dashboards.
country	Adjuster country.	Assignment logic and country workload analysis.
state	Adjuster state/region.	Regional assignment and workload analysis.
specialty	Claim type specialization.	Future assignment logic by claim type.
max_open_claims	Synthetic capacity value.	Workload/capacity dashboards.

payments.csv

Purpose: Settlement/payment records used for financial dashboards.

Column	Meaning	Used for
payment_id	Unique payment identifier.	Primary key for payment records.
claim_id	Claim paid by this payment.	Join to claims.csv.
country	Country copied from claim.	Financial reporting by country.
state	State copied from claim.	Financial reporting by region.
payment_date	Synthetic payment issue/processing date.	Payment cycle time and trend reports.
payment_amount	Calculated payout after deductible and coverage limit.	Paid loss, average settlement, and ROI impact metrics.
payment_status	Issued, Pending, or Cleared.	Payment operations reporting.
payment_method	ACH, Check, or Virtual Card.	Payment channel analysis.

claim_notes.csv

Purpose: Claim history and audit trail for adjuster activity.

Column	Meaning	Used for
note_id	Unique note identifier.	Primary key for claim notes.
claim_id	Claim the note belongs to.	Join to claims.csv.
adjuster_id	Adjuster who owns/entered the note.	Activity by adjuster.
note_date	Synthetic date of note entry.	Claim timeline and audit trail.
note_text	Synthetic adjuster note content.	Claim history and demo narrative.

documents.csv

Purpose: Document metadata that simulates evidence upload and review.

Column	Meaning	Used for
document_id	Unique document metadata identifier.	Primary key for document list.
claim_id	Claim the document belongs to.	Join to claims.csv.
document_type	Police Report, Damage Photos, Repair Estimate, Medical Bill, or Customer Statement.	Evidence/document completeness reporting.
uploaded_date	Synthetic upload date.	Claim timeline and aging.
file_name	Synthetic PDF filename.	Display in document panel; no real files required.

tasks.csv

Purpose: Operational queue showing what work is open, in progress, or done.

Column	Meaning	Used for
task_id	Unique task identifier.	Primary key for task/work queue.

claim_id	Claim the task belongs to.	Join to claims.csv.
assigned_to	Adjuster assigned to the task.	Join to adjusters.csv using adjuster_id.
task_type	Operational task such as verify coverage or fraud review.	Work queue categorization.
task_status	Open, In Progress, or Completed.	Operational status reporting.
due_date	Synthetic due date.	Overdue task and aging dashboards.

3. Dashboard Metrics You Can Derive Directly

Dashboard metric	How to calculate	CSV columns used
Total claims	Count rows in claims.csv.	claims.claim_id
Open claims	Count claims where status is New or Under Review.	claims.claim_status
Paid amount	Sum payment_amount.	payments.payment_amount
Average estimated loss	Average estimated_loss across claims.	claims.estimated_loss
High fraud-risk claims	Count or list claims where fraud_risk_score >= 75.	claims.fraud_risk_score
High severity claims	Count or list claims where severity_score >= 75.	claims.severity_score
Coverage invalid claims	Count claims where coverage_valid is False.	claims.coverage_valid
Claims by country/state	Group claim count by country and state.	claims.country, claims.state
Adjuster workload	Count open claims or tasks by adjuster.	claims.adjuster_id, tasks.assigned_to
Cycle time estimate	Difference between payment_date and claim_date for paid claims.	claims.claim_date, payments.payment_date

4. Recommended Join Keys

Use these joins in Streamlit or pandas:

```
claims.merge(policies, on="policy_id")
claims.merge(adjusters, on="adjuster_id")
claims.merge(payments, on="claim_id", how="left")
claims.merge(claim_notes, on="claim_id", how="left")
claims.merge(documents, on="claim_id", how="left")
claims.merge(tasks, on="claim_id", how="left")
```

Note: These files are synthetic and intentionally simplified for prototype use. They are not actuarial, regulatory, or production-grade insurance data.